

Homework 14 Reflection

Homework 14 Reflection

This Date Display GUI 2 application implements a JavaFX-based date calculator following the MVC pattern. The model stores the selected base date, result date, selected format, and error message using JavaFX properties. The view constructs the UI in Java code, provides a DatePicker, format selector, amount input, direction selectors, unit selector, apply button, and a result section. The controller coordinates the model and view, validating user input, disabling the Apply button when the amount is invalid, and calculating the date result from the selected base date.

Design decisions focused on keeping the MVC structure explicit. The model remains responsible only for state and date calculation logic. The view only builds UI elements and exposes controls. The controller binds the selected date label and result label to model properties through JavaFX bindings, ensuring reactive updates whenever the selected date or date format changes.

A limitation of the environment was that the application could be compiled successfully, but launching the GUI in this terminal session did not produce a persistent visible window. The implementation still follows the handout requirements and is ready to run on a standard desktop with JavaFX support.

JavaFX properties and bindings were applied in the controller: the selected and result date labels use StringBinding to observe both date and format properties, and the Apply button state is bound to validation logic. This ensures the displayed date values remain synchronized automatically with model state changes.